

Counterparts

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Comments welcome

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Chapter 6

Temporal counterpart theory

6.1 Spacetime and the ontology of material objects

Temporal counterpart theory is compatible with a wide range of answers to the question what it is for one thing to be a future or past counterpart of another. However, the reason for being interested in temporal counterpart theory that was stressed in chapter 1—namely, its capacity to reconcile the A-theory of time with the manifold picture—requires us to have some account of how the facts about temporal counterparthood supervene on the facts about spacetime and its fields. And since the things whose temporal counterpart relations we are most interested in are ordinary objects like people, chairs, plants, and planets, this investigation is inseparably bound up with the question what these objects are, and how other kinds of facts about them supervene on the facts about spacetime and its fields.

What would an account of the nature of ordinary objects that was consistent with a spacetime-based fundamental metaphysics look like? One option is “supersubstantivalism”: the claim that ordinary objects *are* certain regions of spacetime. Indeed, it can seem like this is the *only* option: if we think of “spacetime-based fundamental metaphysics” as involving the claim that everything—or every concrete thing—is a spacetime point or a spacetime region, then it will evidently be committed to the view that ordinary objects either do not exist at all, or are not concrete things, or are identical to spacetime points or regions. As I will explain later, I think this way of thinking reflects an unduly restrictive conception of the relation between fundamental and non-fundamental metaphysics. Nevertheless, supersubstantivalism is a serious view, and an investigation of the different forms supersubstantivalism might take will have lessons which will prove useful when we consider alternatives to supersubstantivalism.

To take supersubstantivalism seriously, we will need to face down a host

of elementary arguments against the view (see Hawthorne 2007). The most straightforward of these arguments turn on non-modal, non-temporal predicates which allegedly distinguish ordinary objects from spacetime regions:

No spacetime regions like coffee.

Some people like coffee.

Therefore, some people are not spacetime regions.

Arguments like this one seem to explain the fact that relatively few philosophers profess supersubstantivalism.¹ But as Hawthorne points out, the dialectical force of these arguments is not so great, since it is not clear what can to be said in favour of the central premise that spacetime regions don't have ordinary properties such as that of liking coffee. 'Spacetime region' is not an ordinary concept: it is a concept we learn by means of the role it plays within a certain kind of physical theory. And since the role in question is largely specified using other technical terms of physics, there is no obvious reason why some of the objects which play it should not *also* have the kinds of properties we think ordinary objects have, like liking coffee.

An especially influential class against supersubstantivalism turns on a putative contrast between the modal properties of spacetime regions and those of ordinary objects:

No spacetime region could have had different parts than it actually has.

Some people could have had different parts than they actually have.

Therefore, some people are not spacetime regions.

But again, the source of the justification for the first premise is not clear. It might perhaps be motivated by the thought that the truly fundamental entities are the spacetime *points*, so that the facts about any given spacetime region must supervene on the facts about the points: certainly the most *obvious* way for this to work is for truths about which points are parts of which regions to be necessary. However, the claimed primacy of points is true only on some versions of the spacetime-based fundamental metaphysics; there are plenty of reasons to take seriously ontologies for spacetime physics in which points

¹The following are at least somewhat favourable to supersubstantivalism: Quine 1976, OPW, ***footnote, Field 1989?, Sider 2002, sect. 4.8, Skow 2005***

are not the only fundamental entities, or in which they are not fundamental entities at all. Moreover, once we admit a distinction between 'fundamental' entities and others capable of bearing the required argumentative weight, it is hard to see what would motivate the view that there is only *one* non-fundamental entity corresponding to any given collection of fundamental entities. In general, those who take seriously the contrast between fundamental and non-fundamental entities (or fundamental and non-fundamental interpretations of quantifiers) think that whereas considerations of ontological economy may play some role in guiding our theorising about what there *fundamentally* is, non-fundamental entities in some sense come for free.

The most forceful objections to supersubstantivalism do not turn directly on premises about the non-spacetime-theoretic properties of spacetime regions, but rather proceed by first arguing that supersubstantivalism leads to certain claims of identity among ordinary objects (e.g. the claim that every clay statue is a lump of clay), and then arguing against these claims (e.g. on the grounds that some clay statues are more fragile than any lump of clay). However, to evaluate the first steps of these arguments (from supersubstantivalism to controversial identity claims), we will need to consider more closely the question *which* spacetime regions the supersubstantivalist might identify with ordinary objects.

The best-known form of supersubstantivalism identifies ordinary objects with four-dimensional "spacetime worms": spacetime regions whose extent in the temporal direction is equal to the total duration of the relevant object's existence (as a concrete object). Thus, for example, a person who will die (or did die) at the age of 70 will be identified with a spacetime region such that the longest timelike paths contained within it are around 70 years in length. In a fortunate world where no-one died before 70, the only spacetime regions that are people would be at least this long. This form of supersubstantivalism, however, is of no use to temporal counterpart theorists; it provides far too few ordinary objects. Remember that for temporal counterpart theorists, ordinary objects are the sorts of things that can have properties like *standing*, *having a bath*, and *being thirty years old simpliciter*. And it is easy to see that there is no workable account of these as properties of "long" spacetime worms. Consider *being thirty years old*, for example. If the only people who are thirty years in length are those who will die or did die at the age of thirty, we obviously cannot identify *being thirty years old* with *being thirty years in length*: some people are thirty years old although they will not die for many more years. If spacetime is continuous, then it is plausible that

for every real number x between 0 and 30, everyone who is thirty years old has been x years old, i.e. has a past counterpart who is x years old. So the only versions of supersubstantivalism of interest to temporal counterpart theorists are those according which, if spacetime is continuous, *uncountably many* spacetime regions are people.

A better option for temporal counterpart theorists is a supersubstantialist version of Sider's *stage view* (Sider 1996, 2002). According to this view, the spacetime regions that are ordinary objects are topologically three-dimensional *achronal* regions, i.e. regions such that no point in them lies in the "temporal" spacetime direction from any other point in them. (In Neo-Newtonian spacetime, achronal regions are subregions of simultaneity slices. In relativistic spacetime, achronal regions are *spacelike* regions, those that contain no two points that lie within one another's light cones. For now I will concentrate on the neo-Newtonian case, postponing discussion of the distinctive issues raised by relativistic spacetime until section 6.4.) Where the worm theorist sees one person, the supersubstantialist stage theorist will see continuum many: perhaps all of the maximal achronal parts of the region that is a person according to the worm theorist will be people according to the stage theorist.

The stage view is sometimes described as involving a commitment to the claim that ordinary objects are 'instantaneous'. This way of characterising the view presupposes a certain standard way of mapping the technical vocabulary of spacetime theory (e.g. 'achronal region') onto ordinary temporal expressions like 'instantaneous'. Chapter 4 argued that temporal counterpart theorists should reject this mapping as fundamentally mistaken. Instants of time are nothing like three-dimensional slices through the four-dimensional manifold: the whole four-dimensional manifold exists at each instant, and there is no way in which any given instant privileges part of the manifold. So 'instantaneous' is at best a terribly misleading way of characterising the spatiotemporal shape which, according to the stage theorist, characterises people and other ordinary objects. Stage theorists who insist on the word have to get used to contradictory-sounding claims like 'Some instantaneous things have existed for many years'. Although some have inured themselves to such speeches, I think it is much better to avoid needless paradox-mongering by maintaining a clear separation between ordinary language temporal vocabulary and the technical terminology of spacetime theory.

In general, we should not expect there to be simple recipes for reading

off the ordinary properties and relations of ordinary objects from the physical properties and relations of the regions with which the stage theorist identifies them. The supersubstantialist stage theory does however place an important *constraint* on the answers to questions like ‘What is it to be happy?’, ‘What is it to be red?’, and ‘What is it to be a person?’, namely that it should turn out that achronal spatiotemporal regions can do all these things. But these questions remain difficult even in the presence of the constraint. However, for some ordinary predicates having to do with *geometric* properties and relations, it is clear what the supersubstantialist stage theorist will have to say. For example, one ordinary object (e.g. my arm) is *spatially within* another ordinary object (e.g. me) just in case the former is a *subregion* of the latter. The supersubstantialist stage theorist is thus forced to claim that *no two ordinary objects spatially coincide*: if x and y spatially coincide, then x is spatially within y and y is spatially within x , so x is a subregion of y and y is a subregion of x , so $x = y$. (I take it as central to the theory of spacetime regions that the subregion relation is a partial order). Supersubstantialist stage theorists must thus accept that, for example, a clay statue is identical to any lump of clay which spatially coincides with it, even in cases where we would ordinarily be inclined to think that the statue has only been made a few hours ago while the lump of clay has been around for years.

Sider embraces this commitment to “extreme monism” as a positive feature of the stage theory. He proposes to use the idea that temporal counterparthood is context-sensitive to rebut arguments like the following:

The statue has never been ball-shaped

The lump of clay has been ball-shaped

Therefore, the statue is not the lump of clay

The thought is that ‘has been ball-shaped’ is context-sensitive thanks to context-sensitivity in ‘past counterpart’. In the context of the first premise, we find it natural to resolve this context-sensitivity in such a way that no ball-shaped things count as ‘past counterparts’ of the item referred to by ‘the statue’; in the context of the second premise, we find it natural to resolve it differently, so that certain ball-shaped objects do count among its ‘past counterparts’. There is no single context in which the premises are both true. As I argued in section 4.4, this kind of appeal to the pragmatic effects of word-choice on the resolution of context-sensitivity is a long way from

providing a general device for mimicking the modes of speech characteristic of those who reject the relevant identity claims.

ADD: paragraph arguing that we should embrace those arguments—and multiply de re arguments like ‘The statue used to have parts that were not parts of the lump, so. . .’—and therefore reject stage-theoretic supersubstantivalism unless we really have no better choice.

The stage view is not the only form of supersubstantivalism open to a temporal counterpart theorist. Another option is the *growing worm view*, introduced by Brian Weatherson in an unpublished paper, ‘Growing Individuals and Intrinsic Properties’. This view identifies ordinary objects with *four-dimensional* spacetime regions, while claiming that all *initial segments* of ordinary objects are themselves ordinary objects. A thirty-year old person, for example is a thirty-year-long spacetime region; all of its infinitely many initial segments are also people.²

For the growing worm theorist, the physical facts about the *leading edge* of the spacetime region that is an ordinary object will be especially important in determining whether that object falls under many ordinary predicates, such as shape predicates. For example, it is plausible that *x* is *spatially within y* just in case *x*’s leading edge is a subregion of *y*’s leading edge. Unlike stage-theoretic supersubstantivalism, growing worm supersubstantivalism thus allows for distinct ordinary objects to coincide spatially. For example, a statue which was created this morning might coincide spatially with a distinct lump of clay that has been around for years: we can plausibly identify the statue with a certain worm less than a day long, and the lump with a much longer worm that shares the same leading edge. The growing worm theory naturally suggests the following simple theory of past-counterparthood (for ordinary objects): *all and only the initial segments of an ordinary object are past counterparts of that object*. Given this premise, we can explain the fact that the lump but not the statue used to be ball-shaped by appeal to the fact that the lump but not the statue has a ball-shaped initial segment, where “ball-shaped” is taken to express a property whose instantiation by a given worm depends only on the geometric properties of its leading edge.

²Weatherson’s main argument for the view is that it fits better than the alternatives with our intuitive judgments about the *intrinsicness* of certain properties. For example, *having got taller as one got older* will plausibly be intrinsic according to the growing worm view but not according to the stage view. I wouldn’t want to put much weight on this argument, since I don’t feel a strong enough grip on ‘intrinsic’ to have strong pre-theoretic opinions about which temporal properties are intrinsic.

However, it is hard to see how the growing worm theorist could allow for cases where two distinct objects coincide spatially now and at all past times. Suppose that a certain statue was formed along with the lump of clay that composes it, and will soon be rolled into a ball. Intuitively, the statue is about to be destroyed, whereas the lump of clay is not about to be destroyed but will merely change shape. But in this scenario there are not two distinct worms that look like good candidates to be the statue and the lump, from the growing worm theorist's perspective. Indeed, given the further intuitive judgment that the statue and the lump have *always* coincided, the growing worm theorist seems forced to regard them as identical. For just as it is natural for growing worm theorists to say that the past-counterparts of an ordinary object are its initial segment, so it is natural for them to say that the n -second past-counterpart of an ordinary object is the initial segment derived from it by subtracting its final n -second-long segment. If so, growing worm theorists have to say that for any distinct but coincident x and y , there is some positive n such that x and y did not spatially coincide n seconds ago.

Like the stage-theoretic supersubstantialist, the growing worm supersubstantialist could attempt to appeal to context-sensitivity to dodge the counterintuitive consequences of such identifications. But there is the same reason to think that this move isn't really adequate.

We could also envisage a "shrinking worm" supersubstantialism, which works like the growing worm view in reverse. This will allow for coinciding objects which differ in respect of future-oriented properties like *being about to be destroyed*, while ruling out the possibility that there are objects which have different past careers, but currently coincide and will continue to coincide until they are destroyed.

Can we concoct a version of supersubstantialism that allows for both kinds of cases: objects that coincide now and in the past but not in the future, and objects that coincide now and in the future but not in the past? Yes, at the cost of some artificiality. Essentially, the task is to find a single spacetime region that can serve as proxy for an ordered pair $\langle w, s \rangle$, where w is a full-length spacetime worm and s is a simultaneity slice. Given such a coding, we could develop the view as follows:

- (i) The 3D geometric properties of an object that codes the ordered pair $\langle w, s \rangle$ depend on the 4D geometric properties of the intersection of w and s , and similarly for geometric relations. For example, an object that codes $\langle w, s \rangle$ is spatially within an object that codes $\langle w', s' \rangle$ iff $w \sqcap s$ (the intersection of w and s) is a subregion of $w' \sqcap s'$ (the intersection

of w' and s').

- (ii) Similarly, other ordinary predicates whose application depends on the local character of one's leading edge according to growing worm theorists and of one's trailing edge according to shrinking worm theorists will depend on the local character of $w \sqcup s$.
- (iii) The region coding $\langle w, s \rangle$ is an n -second past counterpart of the region coding $\langle w', s' \rangle$ iff $w = w'$ and s is spatiotemporally earlier than s' by n seconds.

A view of this sort we will be able to allow both for cases where distinct objects coincide in the past and present but not the future, and for cases where distinct objects coincide in the future and present but not the past. We can thus be as liberal in distinguishing temporarily coincident objects as standard (B-theoretic) worm theorists.

There are various ways to actually do the coding. For example, we could take the object coding $\langle w, s \rangle$ to be the "gappy worm" $w \setminus s$, i.e. the maximal subregion of w that does not overlap s . Or we could take it to be the "supplemented worm" $w \sqcup s$, the minimal region that has both w and s as subregions. By requiring the relevant w to be topologically well behaved (e.g. a bounded open region, or a bounded closed region that is the closure of some open region), both of these proposals allow us to straightforwardly recover the ordered pair $\langle w, s \rangle$ from the region that represents it.

Proposals like these face a worry about arbitrariness that makes them hard to take fully seriously. But as usual, such worries can be shrugged off by means of a judicious invocation of vagueness-related ideology. It is, indeed, implausible that either 'People are gappy worms' or 'People are supplemented worms' is *definitely true*. But perhaps this just goes to show that there is a range of proposals whose disjunction is definitely true, although none of the disjuncts are. The literature on vagueness features a disparate range of foundational proposals about what this talk of 'definiteness' might come to, which share the key feature of allowing for disjunctions to be definitely true even when none of their disjuncts are. These proposals all, in one way or another, provide the resources to explain why we should not be put off from accepting certain disjunctions by the concern that the choice among the disjuncts is arbitrary.

Anyhow, it is not even *true* that there are no non-arbitrary grounds for choosing between the two coding-based theories we have considered.

For unlike the “gappy worm” theory, the “supplemented worm” theory provides a natural coding not only for the ordered pairs $\langle w, s \rangle$ where w intersects s , but also for those where w fails to intersect s . The latter regions are natural candidates to be the “non-concrete” objects which—as we saw in section 3.6—temporal counterpart theorists have good reason to posit among the temporal counterparts of objects that undergo creation and destruction. The thought is that while you now fuse of a certain worm w and a certain simultaneity slice s which intersects w , after you die you will fuse the very same worm w with other simultaneity slices which fail to intersect w . Of course, in the standard philosophical sense of ‘concrete’, the fusion of a worm with a simultaneity slice that does not intersect it is just as concrete as any other spatiotemporal region. But this is not the sense of ‘concrete’ that featured in the argument that ordinary objects have non-concrete temporal counterparts. For the purposes of that argument, what matters is just that the things we are calling ‘non-concrete’ do not fall under everyday, positive, changeable, non-modal, non-temporal predicates—predicates like ‘overweight’, ‘conscious’ and ‘in Oxford’. And since it is part of the proposal that the applicability of predicates like these to $w \sqcup s$ mostly depends on the local properties of the intersection $w \sqcap s$, the proposal naturally leads to the conclusion that these predicates fail to apply in any non-trivial way to $w \sqcup s$ when w does not intersect s at all. By contrast, temporal predicates like ‘was once a philosopher’ can still apply in that case—on the proposed account of past-counterparthood, $w \sqcup s$ was once a philosopher iff for some slice s' spatiotemporally earlier than s , $w \sqcup s'$ is a philosopher.

Many philosophers regard identifications like ‘Some people are spacetime regions’, which bridge what would appear to be different ontological categories, as fundamentally misguided, and are unmoved the alleged advantages in “ontological economy” that such identifications allow. We could call these philosophers ‘Butlerians’, in honour of Bishop Butler’s delightful maxim ‘Each thing is what it is, and not another thing’ (*cite*). If the manifold picture said that *every* concrete entity is a spacetime point or region (where the relevant interpretation of ‘every’ is dual to a sense of ‘some’ in which it is obvious that some things are people) then it would not be something Butlerians could take seriously. But the manifold picture does not have to be understood like that. The key idea—that “fundamentally speaking”, the only things there are are spacetime points and/or regions—is compatible with the thought that there are lots of concrete objects besides spacetime

points and regions (in some perfectly good sense of 'there are').

How should Butlerian temporal counterpart theorists who endorse the manifold picture approach the question what it is for one ordinary object to be a future- or past-counterpart of another? The central challenge they face derives from *Manifold Supervenience*, the claim that all truths (both qualitative and haecceitistic truths), follow with metaphysical necessity from the truths about physical fields in spacetime (including haecceitistic facts about the values of physical fields at particular spacetime points), together with an appropriate 'that's all' claim. *Manifold Supervenience* is a central component of the manifold picture as developed in ???. Given *Manifold Supervenience* and the eternity of metaphysically necessary truths (which I will here take for granted), there can be no change in ordinary objects without change in the spacetime-theoretic properties of spacetime points or regions. Since metaphysically necessary truths are always true, and hence true "at" all future- and past-counterpairings, the action of a future- or past-counterpairing on spacetime points and regions determines its action on everything else. So even though we deny that ordinary objects are spacetime points or regions, we can expect an answer to our question what it is for one ordinary object to be a future- or past-counterpart of another to flow from an answer to the question what it is for a spacetime point or region to be a future- or past-counterpart of another, together with a general account of how the facts about any given ordinary object supervene on facts about certain spacetime points or regions.

The supersubstantialist views discussed above depend crucially on our including spacetime regions as well as points in the fundamental ontology, and on our willingness to reject mereological essentialism as regards spacetime points and regions. Given these commitments, supersubstantialists are free to hold that *points* are unchanging—i.e. that every temporal counterpairing maps every point to itself. Unlike supersubstantialists, Butlerians are free to take seriously versions of the manifold picture which include only points in the fundamental ontology, or which endorse mereological essentialism for regions. Let us focus on the points-only version of the view, since this greatly limits the options for developing a plausible account of change at the ground-floor level. In certain spacetime structures, there would be no problem. For example, there is a structure called 'Newtonian spacetime', which supplements Neo-Newtonian spacetime with a natural relation partitioning the set of all points into family of parallel timelike trajectories—a distinguished criterion of absolute motion and rest. In that setting, it is

natural to suppose that the temporal counterparts of any given spacetime points are just those spacetime points that belong to the same “absolute rest” trajectory as it, and that each counterpairing acts on spacetime points simply by shifting them forwards or backwards by a constant amount on these trajectories. But without this structure to guide us, we face a problem of arbitrariness. In Neo-Newtonian spacetime we can impose some fairly natural constraints: each temporal counterpairing should act on the set points comprising a simultaneity slice (or flat spacelike hypersurface) by shifting it forward or backward by a uniform amount; the lines generated by taking a given point together with all of its temporal counterparts should all be straight timelike lines, parallel to one another. But without a criterion of absolute motion and rest, we have no non-arbitrary way to decide which *direction* the lines in question should point in, or how fast any given object is moving relative to the frame they pick out. (The situation is similar in Minkowski spacetime, but this case raises certain complexities best postponed until section 6.4.)

But maybe that isn’t so bad. As in other such cases, we could just play the vagueness card, declaring that there is no *definite* answer to the question ‘In what timelike direction do the lines connecting spacetime point to its temporal counterparts point?’. The worry about this move is that we don’t want the vagueness in the question ‘What will this spacetime point be like in the future?’ to spill upwards into the macroscopic realm—for example, under normal circumstances we don’t want there to be any significant vagueness in the question ‘What will this planet be like in the future, in physical respects?’. We want to be able to say without qualification that the planet is travelling in a certain spatial direction without taking a stand on how that direction relates to the set of trajectories that map spacetime points to their temporal counterparts. This might seem hard to manage, given that *ex hypothesi*, the facts about what the planet is doing *supervene* on the facts about what the spacetime points are doing. But on reflection there is no real problem here, given that the supervenience of one subject matter on another does not mean that *any* difference with respect to the latter subject matter entails some difference with respect to the former. There are many maximal possible states of affairs which are accurate as regards the qualitative facts about spacetime while differing as regards which spacetime points play which roles. Among these states of affairs, there is none that of which the predicate ‘will obtain *n* seconds from now’ is *definitely* true. Rather, for each *n*, there is a set of many states of affairs, each of

which is a borderline case of ‘will obtain n second from now’. However, there is no obvious reason why there should be any difference among the states of affairs in these sets as regards what they entail about the qualitative properties of and relations among particular planets (their masses, relative speeds, relative distance, etc.).

So, what sort of answer can we expect Butlerians give to questions about what it is for a given ordinary object to be a certain way? What we ultimately want out of the theory of ordinary objects is an account of the mapping from propositions about ordinary objects (both *de re* propositions about particular ordinary objects, and propositions involving quantification over ordinary objects) to the spacetime-theoretic propositions upon which they supervene. (We might want to add, further, that relations of *grounding* or *reduction* or *identity* hold between the ordinary-object propositions and the spacetime-theoretic propositions to which they are mapped.) Of course, providing a *complete* such account would require answering all the distinctive questions of the philosophy of mind, and ethics, and most of the rest of philosophy. What we can reasonably hope for in the present context is an account of some especially tractable propositions about ordinary objects—e.g. concerning their shapes and spatial relations. Our survey of forms of supersubstantivalism suggests a plausible strategy for accounting for these propositions, according to which the relevant properties of an ordinary object o are “inherited” from a certain collection of spacetime points with which o is in some way intimately associated:

- (i) o has shape-property $P \Leftrightarrow o$ is associated with some spacetime points whose sum (least upper bound in the partial order of subregions) is an achronal spacetime region that has P .
- (ii) $o_1 \dots o_n$ stand in spatial relation $S \Leftrightarrow$ there are achronal spacetime regions $r_1 \dots r_n$ that stand in S , such that r_1 is the sum of the spacetime points associated with $o_1 \dots$ and r_n is the sum of the spacetime points associated with o_n .

What can be said about the relation of “association”? On one view of ordinary objects, ordinary objects have fundamental objects as *parts*, and the kinds of spatial properties we are concerned with are ones that depend on the spatial properties of and relations among an object’s parts.³ On this kind of approach, if the only fundamental objects are spacetime points, we need to

³*cite Goodman, ‘Matter and Mereology’*

say that ordinary objects have spacetime points as parts; and the natural way of understanding (i) and (ii) is to take the region “associated” with an object to be the minimal region having all the same spacetime points as parts. Alternatively, we might reserve the predicate ‘part’ for relations between objects of a more familiar sort, and leave ‘associated with’ as a primitive of the theory of non-fundamental objects—we no more expect an informative general account of what it is for a spacetime region to be associated with an arbitrary ordinary object than an informative general account of what it is for something to instantiate an arbitrary property. Just as it is a basic fact about the particular property *sphericity* that it is necessarily instantiated by all and only the spherical things, we might think of it as a basic fact about a particular ordinary object—say, *Sputnik*—that it is necessarily associated with whichever region uniquely satisfies a certain condition Φ . In describing any domain of non-fundamental entities, we are going to have to allow ourselves certain unanalysed predicates which capture the “structure” of the domain in question: membership for sets, instantiation for properties, truth for propositions. . . . It would not be terrible to add *association* for ordinary objects to this list.⁴

Whether we identify association with parthood or take it as primitive, the present conception makes ordinary objects in many ways like *properties of spacetime points*—properties with the special feature that always (and necessarily?) all of their instances belong to the same achronal region. Each ordinary object corresponds to one such property. And just as *de re* propositions about a particular property—e.g. the proposition that sphericity has at least three instances—supervene on (are grounded in, reduce to. . .) propositions which do not involve reference to properties—e.g. the proposition that there are at least three spheres—so, propositions about a particular ordinary object—e.g. the proposition that *Sputnik* is spherical—supervene

⁴I could have chosen the label ‘occupation’ rather than the bland ‘association’ for the special relation between ordinary objects and spacetime points. But there are also other relations between ordinary objects and spacetime points for which we might want to call ‘occupation’: for example, there is the relation *p is associated with a temporal counterpart of o*. And if we want to speak of occupation of *regions* as well as points, there are further possibilities, such as *r is the fusion of all the points associated with temporal counterparts of o* and *r is the fusion of all the regions associated with temporal counterparts of o*. I can’t see any substantive distinction between these different ways of using the term ‘occupation’. But since some philosophers ***cite Gilmore, Hudson, . . . *** think that ‘occupation’ can be used to pick out a relation between ordinary objects and spacetime regions in such a way as to leave room for substantive debate as regards such questions as whether ordinary objects only occupy one spacetime region or many, I prefer to avoid the term.

on propositions which do not involve reference to ordinary objects, e.g. the proposition that the Φ region is spherical. Of course, most properties of spacetime points will not correspond to any ordinary object. For example, the property *having mass-density greater than n* does not—there is certainly no *ordinary* object such that it is necessary that it has the same shape as the sum of all points with mass-density greater than n . In our account of what it is to be an ordinary object of a particular *kind*—a person, or a body, or a planet, or a work of art—we will want to look at specific further features of the property corresponding to the object in question.

(If we thought we needed to allow for cases of *necessary* spatial coincidence among ordinary objects, we might find we needed some other, more discriminating account of the “structuring relations” for ordinary objects. Using these relations, we could both say something substantive about what the association relation consists in, and also give an account of whatever distinctions we take there to be between necessarily-coincident ordinary objects. The literature contains several putative examples of necessary coincidence. For example, Fine (2000a) imagines a case where someone contrives to write two different letters, in two different languages, using the same inscription—the letters are different in various ways (e.g. they have different addressees, different contents, different styles) but, according to Fine, it is necessary that if either exist, both exist and are coincident. One possible scheme for admitting such possibilities is to take as basic the relation between an ordinary object and the property of regions to which it “corresponds”, while adopting a conception of properties on which properties can be distinct but necessarily coextensive—the letters will then correspond to distinct coextensive properties. Another option is to introduce some kind of hierarchical structure within the domain of ordinary objects: the first-level objects will correspond to properties of spacetime regions, while second-level objects will correspond to properties of first-level objects, etc. (Relevant here are the pictures of ordinary objects described in Fine 2000b and Johnston 2006 and *Goodman*.) For now, however, I will not try to speculate on the best way to introduce such additional fine-grainedness.)

Even while focusing on ordinary objects in general rather than particular kinds of ordinary objects, we can say a bit more about the kinds of properties of regions to which ordinary objects correspond. First, it must be a *temporary* and (hence also) *contingent* matter which spacetime points are associated with a given ordinary object. For given that we are not in the grip of the impulse towards ontological reduction which motivates supersubstant-

ivalism, we presumably want to allow for objects which temporarily and contingently spatially coincide. Since coincidence involves being associated with the same points, it must be true in any case of contingent coincidence that at least one of the coinciding objects could have been associated with different points; and in any case of temporary coincidence, that at least one of the coinciding objects was or will be associated with different points.

Second, the properties to which ordinary objects correspond cannot (in general) be purely *qualitative* properties of points. For almost all ordinary objects change their shapes over time. Suppose that an object o having a shape-property P corresponded to a qualitative property Q of spacetime points, i.e. that necessarily, o is associated with whichever points have Q . Then the sum of the Q points must have P . But the proposition that the sum of the Q points has P is purely qualitative: so for the temporal counterpart theorist who holds that all change is *de re* change, it must be *always* true, and hence it is always the case that o has P . For o to undergo change of shape, the condition under which a point is associated with o must be non-qualitative—it must be the sort of condition such that specifying it spacetime-theoretic terms would require reference to some specific spacetime points.⁵

The most obvious candidates to be the non-qualitative properties of points corresponding to ordinary objects would be properties of the form *being one of spacetime points p_1, p_2, p_3 which is such that $\phi(p_1, p_2, p_3 \dots)$* . On the conception of association as parthood, this would correspond to an object which necessarily has whatever point-parts it has if it is concrete at all, but would fail to be concrete unless the points satisfy the relevant condition ϕ . While this doesn't work, it is not as much of a non-starter as it may initially look. While it is obvious that we gain and lose parts, it is not at all obvious that any of the parts that we gain and lose are spacetime points, since it is not obvious that we have such parts in the first place. The fact that we move around doesn't settle the matter either. For all we have said, the spacetime points that are our parts may move around with us: for example, getting closer to the spacetime points that are part of other people as we get closer to those people. This might correspond to a temporal counterpart relation on which the temporal counterparts of a given spacetime point are traced by

⁵Note that this form of argument works not just for shapes but for any other qualitative property which is always instantiated by both or neither of any pair of ordinary objects associated with the same points, e.g. extrinsic properties like 'being in the same room as a statue'. It is safe to say that every ordinary object undergoes qualitative change with respect to some such properties.

following the flow lines of some the matter fields through that point, so that when the matter clumps together, spacetime points that are currently further apart will be closer together. This conflicts with the earlier suggestion that the set of temporal counterparts of any given spacetime point compose a straight timelike line; but that was just a natural suggestion, not the conclusion of an argument. The reason that we need to allow for changes in the set of spacetime points associated with an object is that there are cases of temporary coincidence between objects which sometimes fail to coincide while still being concrete. For example, a statue may coincide with a certain lump of clay, even though later more clay will be added to enlarge the statue, at which point it will be spatially larger than the lump. Given (ii), this means that the statue and the lump are associated with the same spacetime points, although later it will be the case that some spacetime points are associated with the statue but not the lump. So, the non-qualitative properties corresponding to ordinary objects need to be more flexible ones—more along the lines of *being one of $s_1, s_2, \dots, s_n$ and such that ϕ , or being one of $s'_1, s'_2, \dots, s'_m$ and such that ϕ' , or...* Given that we are going to need this kind of flexibility in any case, there is no strong reason not to stick with the earlier idea on which ‘the temporal counterparts of any point form a straight line’ is definitely true although the extension of ‘temporal counterpart’ is so vague that any two points not in the same simultaneity slice are a borderline case of ‘temporal counterpart’. This idea entails that insofar as the temporal profile of an ordinary object is not vague, it is invariant under spatial shifts in the underlying spacetime points, in the sense that if o is associated with a set S of points in possible situation w , and w' differs from w just by a spatial shift π such that every point p plays in w' the qualitative role that $\pi(p)$ plays in w , then o is associated with $\pi(S)$ in w . Even though regions (including spacelike slices) are not in the fundamental ontology, the properties corresponding to ordinary objects could still be sensitive only to those shifts in the qualitative roles of points that also shift the qualitative roles of spacelike slices.

6.2 Fission and the linearity of time

Standard tense logic includes the axiom schemas of *past totality* and *future totality*:

$$\begin{aligned} (P\phi \wedge P\psi) &\rightarrow (P(\phi \wedge P\psi) \vee P(\phi \wedge \psi) \vee P(\psi \wedge P\phi)) \\ (F\phi \wedge F\psi) &\rightarrow (F(\phi \wedge F\psi) \vee F(\phi \wedge \psi) \vee F(\psi \wedge F\phi)) \end{aligned}$$

For example: if Bill will be rich and will be happy, then either Bill will be rich before being happy, or he will be rich and happy simultaneously, or he will be happy before being rich. This is plausible. If we are told that Bill will be rich and will be happy but will never be both rich and happy, we feel licensed to ask 'Which will he be first?'; we will be completely stumped by the answer 'Neither'.

In the standard model theory for tense logic, these schemas correspond, respectively, to the following claims about the structure of the before/after relation $<$ on the domain of instants:

- (i) If $t_1 < t_3$ and $t_2 < t_3$, then either $t_1 < t_2$ or $t_1 = t_2$ or $t_2 < t_1$.
- (ii) If $t_3 < t_1$ and $t_3 < t_2$, then either $t_1 < t_2$ or $t_1 = t_2$ or $t_2 < t_1$.

Structures with Past Totality but not Future Totality thus correspond to "branching" time-orderings. This has been taken seriously as a way of representing the elusive claim that the future is "open". But (as I argued in passing in section 1.3), rejecting Future Totality does not really answer to the intuitions behind that claim, since as it involves saying that 'There will be a sea battle' is just *true* at times before the branch.

In the course of introducing temporal counterpart theory, Sider offers a new argument against Future Totality, which turns on the fission cases familiar from the literature in personal identity (e.g. Parfit 1971). In a typical such case, a person—a rather special person, whose brain exhibits an unusual degree of bilateral symmetry and redundancy—has just walked into a doctor's office. The doctor will first destroy all of the person's body except for the brain, then bisect the brain, and transplant each of the halves into a new body, with the result that after the transplant, each of these bodies will exhibit many of the distinctive dispositions that the body now standing in the doctor's office currently exhibits. Let's suppose further that things will work out so that the two recipient bodies have careers of different sorts: e.g. one will have the features that would in a normal case make it the body of a rich person, while the other will have the features that would in a normal case make it the body of a poor person. The puzzle is: what is going to happen to the person—call him Fred—now standing in the doctor's office? Will Fred be rich? Will Fred be poor?

There is a standard menu of options:

- (i) (Parfit 1971): Fred will never be rich, and will never be poor. In fact he will soon cease to exist. There will be a rich person and a poor person

who are like Fred in certain respects, but neither of these people will *be* Fred.

- (ii) Fred will be rich *and* poor. He will be in two places at once, thinking two sets of thoughts, and unable to introspectively access his total package of experiences.
- (iii) (Williamson 1990): Either Fred will be rich and will never be poor, or else he will be poor and will never be rich. In addition to Fred, there will one other person around who seems to remember Fred's childhood, etc., but in fact came into existence at the time of the operation.
- (iv) (Lewis 1976a): The question has a false presupposition. Strictly speaking, there are (at least) *two* people now standing in the doctor's office. So far, they have always coincided. But they will soon cease to coincide. One of them them will later be rich, and one of them will later be poor. No new people will be created when the operation is performed.

The obvious worry about (iii) is that each of its disjuncts involves an obnoxious kind of metaphysical arbitrariness, which cannot be tolerated if we don't believe in 'soul pellets' or other such special-purpose additions to fundamental ontology. And (once again), we can respond to this worry by invoking vagueness-related ideology: 'Fred will be rich' and 'Fred will be poor' are both "indefinite", even though their disjunction is definitely true and their conjunction is definitely false. (And this is indeed the approach taken by Williamson 1990.)

None of these options is easy to believe: that's why the case is a puzzle. (Note that even if we see no general problem with temporary coincidence, e.g. between statues and lumps of clay, we might still find it extremely hard to believe that distinct *people* can coincide, as they do on view (iv).) Thus it is of great interest if it turns out that (i)–(iv) are not the only options. One of Sider's arguments for temporal counterpart theory is that it opens up a new option, which he thinks more attractive than the standard ones:

- (v) Fred will be rich. Also, Fred will be poor. But Fred will never be *both rich and poor*. Nor will he ever be in two places at once, or unusually lacking in introspective access to his total package of experiences. He will always be a psychologically ordinary, spatially unified person.

Given temporal counterpart theory, option (v) amounts to the claim that Fred has rich people and poor people among his future counterparts, but

has no temporal counterparts who are both rich and poor, or in two places at once. We will get this result if, like Sider, we identify people with stages, and identify the counterpart relation with the "I-relation": the relation such that, according to Lewis, people are maximal aggregates of mutually I-related stages.

Sider's proposal requires giving up Future Totality. For given Future Totality, 'Fred will be rich and Fred will be poor, but Fred will never be both rich and poor' entails 'Either Fred will be (a rich person who is going to be poor), or Fred will be (a poor person who is going to be rich)'. And both disjuncts of this disjunction are obviously false. Given our stipulations about what is going to happen, there is no reason at all to think that Fred's rich future counterparts have poor future counterparts, or that his poor future counterparts have rich future counterparts.

I expressed the puzzle as one about what *will* happen to the person who is about to undergo fission. But we can also consider a backward-looking version of the puzzle, where we consider (what seem to be) the two people who emerge from the fission operation—rich Ed and poor Ted—and ask what each of them has been like up to now. The standard options yield the following answers: (i) Both Ed and Ted have only recently been created; (ii) The question has a false presupposition, since really only one person emerged from the operation: Ed is Ted; (iii) One of the two has lived a long life, while the other was only recently created; and (iv) Both have lived long lives, but until recently they were coincident. What should we say if we favour Sider's answer (v) to the question about Fred? Well, if Ed and Ted are both future counterparts of Fred, then each of them has Fred as a past counterpart. So both Ed and Ted are people who have once stood in a doctor's office; more generally, when Ed and Ted consult their apparent memories and say 'I was once a child who did this and that', they speak the truth. In this respect, then, Sider's account of the case is in agreement with Lewis's (iv). Must Sider then agree with Lewis that Ed and Ted were once in exactly the same place, with exactly the same parts, etc.? That claim is *prima facie* in conflict with the claim that there really was just one person standing in the doctor's office.

Sider's way to resolve this conflict is to deny the permanence of distinctness. It is true that Ed and Ted *were* coincident, but when they were coincident, they were *identical*.⁶ There have never been *two* coincident people.

⁶Note that the mere claim that Ed and Ted share a past counterpart does not, on either of the counterpart-theoretic accounts of the multiply *de re* which we found to be adequate,

Given some standard tense-logical principles that Sider does not dispute, the failure of the principle that distinctness is permanent entails the failure of the principle that identity is permanent. (From $x \neq y \wedge Px = y$ we infer $HFx \neq y \wedge Px = y$ by Mixing, and hence $P(x = y \wedge Fx \neq y)$: a past failure of the permanence of identity.) I have already done as much as I can (in section 4.4) to explain why I think we should hold onto the necessity and permanence of identity, whether or not we are counterpart theorists.

However, there is another option for Sider which is compatible with the permanence of identity and distinctness. On this approach, we affirm that Ed was once a child and that Ted was once a child, but deny that it has ever been true that (Ed and Ted are both children). This will require a failure of Past Totality that matches the contemplated failure of Future Totality, given that it has obviously never been the case that (Ted is a child and Ed is a former child) or (Ed is a child and Ted is a former child). In fact, given the results of section 4.3, it would not be surprising for failures of Past Totality to crop up in tandem with failures of Future Totality. For as we saw there, Future Totality is equivalent to Past Totality in the framework of COUNTERPAIRINGS, given the stipulation that future and past counterpartings are one another's inverses: this was one of the few logical upshots of COUNTERPAIRINGS that did not correspond straightforwardly to any feature of the standard model theory.

(Of course, Sider is already committed to failures of Past Totality in cases of *fusion*, which work like fusion cases in reverse. This is distinct from the failure of Past Totality we are currently contemplating, although the fact that fusion cases motivate the rejection of Past Totality to the same extent that fission cases motivate the rejection of Future Totality does disarm one objection that someone might have to the suggestion.)

But what was Ted like when Ed was a child, if he wasn't a child himself? The only remotely plausible thing to say is that Ted was *non-concrete* whenever Ed was a child. And likewise, Ed was non-concrete whenever Ted was a child. In the counterpartings framework, we can picture the scenario as follows: there are three relevant people, Fred, Ed, and Ted, and one relevant non-concrete future person, call it Nonc. There are two relevant past-counterpairings: R_1 maps Ed to Fred and Ted to Nonc, and R_2 Ted to Fred and Ed to Nonc. And there are two relevant past-counterpairings, namely the converses of these future counterpartings: R_1^{-1} maps Fred to Ed

entail that Ed and Ted were once identical.

and Nonc to Ted, while R_2^{-1} maps Fred to Ted and Nonc to Ed. So, while we get some natural speeches like 'I was once a child' to be true in Ed's and Ted's mouths, we also get some weird speeches to be true, like 'I was once such that I had never been a child but would later be an adult'. While this is strange, remember that the basic idea of (v) already commits us to accepting some quite odd speeches from Ed and Ted—for example, 'I am not, never will be and never have been rich, but I was once such that I would later be rich'. And if we develop the view in Sider's way, rejecting the permanence of distinctness, we of course get a further range of oddities. Among the two versions of option (v), my preference is for the one that keeps the necessity of identity and rejects Past Totality in cases of fission; but this is driven primarily by my belief that the necessity of identity is an especially fundamental logical principle, whose rejection would give an implausibly "orthographic" cast to our account of the meaning of the tense operators, whereas the rejection of Past Totality seems like no further logical cost once one has already given up on Future Totality.

It's worth being clear that option (v) (in either version) need not be the sole preserve of the temporal counterpart theorist: the position is one that can be articulated by anyone who can make sense of tense operators. If one thought of temporal counterpart theory as crucially involving the denial that instants of time play the standard instant role, one might regard the combination of (v) with temporal counterpart theory as more plausible than the combination of (v) with the negation of temporal counterpart theory, on the grounds that if instants play the standard role, (v) would commit us to the implausible thesis that there are future instants at which Fred is rich and future instants at which Fred is poor, although none of the former are before or after any of the latter. But I don't think that this is a good way to think about temporal counterpart theory—rather, everyone should do their best to find an interpretation of 'instant' that plays the standard role; and if you like the view that every instant is either before or after every other, you should to that extent be unsympathetic to (v).

To conclude: while it's good to get new ways out of old puzzles on the table, my considered view is that whether or not one is a temporal counterpart theorist, one should prefer one of the old solutions (i)–(iv) to the new solution (v). I'm especially fond of solution (iii), on the grounds that the central objection people always bring up to this proposal—namely that it is unacceptably arbitrary—seems to be nothing more than one of the perennial surfacings of a kind of concern which we should expect the correct account

of vagueness to set to rest. Solution (iii) seems to do better than solution (i) at vindicating the idea that the difference between cases where one ceases to exist and cases where one continues to exist should be something that matters, something that it makes sense to care about; and indeed, despite the fact that Lewis and Sider both give great weight to these broadly ethical considerations in defending their own solutions, it's not at all clear that any of the other views does better than (iii) in this regard.

6.3 Time travel

Ted: Hey! That's us. We're back in San Dimas.

Bill: Yeah, only now it's not now. It's last night.

Ted: Dude.

Bill: What?

Ted: Let's go talk to ourselves.

Bill: Excellent.

Ted: What'll we say anyway?

Bill: I dunno—let's go find out.

(Flash Forward)

(The other Bill and Ted are now Bill2 and Ted2. Our Bill and Ted are just Bill and Ted.)

Bill & Ted: 69, dudes!

Bill2 & Ted2: Whoa!

Bill: Look dudes, we gotta go.

Ted: Rufus!

Bill: Listen to this dude Rufus, he knows what he's talking about.

Ted: Right. Oh, and Ted, give my love to the Princesses.

Ted2: Who?

Ted: You'll see.

Rufus: Gentlemen, is everything all right?

Bill: Yeah, except how come the number for San Dimas brought us here instead of to tomorrow, Rufus.

Rufus: Because in San Dimas it is tomorrow, William. You have to dial one number higher.

Bill: Oh yeah, thanks Rufus.

Rufus: And you better hurry, because you don't have much time left.

Ted: What do you mean, Rufus? We got ten hours left.

Rufus: Uh-uh, you got 2 hours left.

Bill: Ted you forgot to wind your watch again. Even after you reminded yourself not to.

Ted: Well I better remind myself again. (to Ted2) Ted, don't forget to wind your watch! (Ted2 nods)

Bill & Ted: Thanks Rufus. Catch ya later Bill and Ted!

Ted: That conversation made more sense this time.

Bill and Ted's Excellent Adventure

Temporal counterpart theory, as we have developed it, sheds a new and unfamiliar light on debates about the possibility of time travel. So as to postpone discussion of the distinctive issues raised by Relativity, let us consider a frankly science-fictional form of (putative) time travel, implemented

by means of weird laws of nature in a Neo-Newtonian spacetime. Let me describe one form these weird laws might take, in spacetime-theoretic terms. There is a certain very natural symmetric relation which holds between certain pairs of three-dimensional achronal regions: when two regions stand in that relation, we say they 'form a portal'. It is a law of nature that any two regions which form a portal are of the same shape and size. When two regions form a portal, the disposition of physical fields immediately *futurewards* of one of them fits with the disposition of physical fields immediately *pastwards* from the other, in the same way in which conditions immediately futurewards of any ordinary region fit with conditions immediately pastwards of that same region. Thus, for example, since the laws require field values to be *continuous* at ordinary regions, so that the limiting value of a field as one approaches a point from the future direction must equal its limiting value as one approaches that point from the past direction, the laws require that when R_1 and R_2 form a portal, the limiting value of a field as one approaches a point on R_1 from the future direction must equal its the limiting value as one approaches the corresponding point on R_2 from the pastward direction.⁷ So, if conditions immediately pastwards from R_2 are such as to constitute the existence of a person, conditions immediately futurewards from R_1 will be such as to constitute the existence of a very similar person.

Since this description of the relevant worlds did not employ any tense operators, temporal counterpart theorists have just as much reason as anyone else to regard it as a genuine possibility. But this is not yet to settle the further question whether a world like that would be a world in which things undergo *time travel*. What would it mean to answer this question affirmatively? At the very least, it would require claiming that when two regions form a portal, objects associated with spacetime regions immediately futurewards of one can be *future counterparts* of objects associated with spacetime regions immediately pastwards of the other. By allowing considerations of what Lewis calls 'personal time' (1976b) to govern the assignment of temporal counterparts, we will, for example, be able to endorse speeches like 'I will soon be gazing at a dinosaur' when the region of spacetime with which the speaker is associated is immediately pastward from one member of a portal whose other member is immediately pastward from the region associated

⁷To make sense of this talk of 'corresponding' points, suppose that when R_1 forms a portal with R_2 , each part of R_1 forms a portal with some geometrically indiscernible part of R_2 .

with a person who is gazing at a dinosaur.

Given this natural way of assigning temporal counterparts, then, these physical laws make possible a surprising kind of 'spacetime travel', wherein one suddenly comes to be associated with (or 'occupy') spacetime regions that lie far, in the pastward or futureward direction, from the spacetime region with which one is currently associated. We should be wary of calling this 'time travel', however, bearing in mind chapter 4's rejection of the identification of instants of time with regions of spacetime. And indeed, there are several important differences between what we get from temporal counterpart theory and the standard, B-theoretic understanding of time travel we get from Lewis's classic 'The Paradoxes of Time Travel' (1976b).

(i) Even if I step into one end of a portal whose other end is just a few years back, I am not going to be able to visit *myself*. At best, I will visit one of my past counterparts. And my past counterparts are not me. When outsiders overhear us and say 'There are two people having a very strange conversation', they will be speaking the truth. And I can speak truly when I say things like, 'You are shorter and younger and less knowledgeable than I am' to my past counterpart. By contrast, from a B-theoretic standpoint, there is only one person involved in the relevant conversation, and some sort of very special pleading would be required to vindicate claims like 'You are shorter than I am'.

Overall, I think this is a significant benefit of the temporal counterpart theorist's account of the situation. The temptation to think of the relevant encounters as involving two distinct people is extremely hard to resist. (This is especially true if we imagine a case where I and 'my former self' settle into a long-term friendship or rivalry.) People in such cases will endorse a wide variety of claims of the form 'This person [is taller than/is older than/is shooting at] that person, and not *vice versa*'. If 'this person' and 'that person' both refer to the same person, it is very hard to see how to make charitable sense of such claims (cf. Fine 2003). The most plausible avenue for the B-theorist in such cases is to appeal to a hypothesis of 'referential shift', whereby in context 'this person' and 'that person' have distinct referents—items which do not fall under the standard extension of 'person', but do count as 'people' relative to the unusual standards that prevail in the context of discourse about the time traveller. For example, it might be suggested that 'this person' and 'that person' refer to three-dimensional stages, or to longer proper parts of the worm that belongs to the standard extension of 'person'. The problem with such suggestions is that the suggested substitute referents

do not, under a B-theoretic analysis, have the properties we are inclined to attribute to them. We might want to say, for example, that 'This person is twenty years older than that person', or 'This person will someday invent a kind of time travel, whereas that person's inventing days are done'. But on a standard B-theoretic understanding of temporal predicates like 'twenty years older than' and 'will invent a kind of time travel', they will not apply to stages, or to the short-lived worms which might be suggested as substitute referents. It might be proposed that in context, the extensions of all such predicates shift in tandem with that of 'person' so as to render true what we are inclined to say about 'this person' and 'that person'. But what is the alternative semantic account of these predicates that prevails in the context? It begins to seem like it must be some form of temporal counterpart theory! But given the understanding we have developed of how temporal counterpart theory has to work, it is utterly implausible to suppose that there are some special contexts for which temporal counterpart theory is correct, and others for which the standard B-theory is correct. The gulf between the two views is far too wide for such a compromise to be tenable. They require completely different accounts of what kinds of things instants of time *are*, and completely different accounts of what it is to be *F* 'at' an instant of time, for pretty much every predicate in the language.

(ii) For the same reason that travel by portal does not allow us to visit *ourselves*, it does not allow us to visit our friends and relations. The best we can hope for is to meet their temporal counterparts. For example, if I travel through a portal whose other end lies in the futurewards direction, I will not be able to visit my children. I can visit their adult future counterparts, but I am not these people's father: their father is one of *my* future counterparts. And if I ever want to see my own children again, I will have to travel back through a portal whose two ends are *exactly* as far apart (in the temporal direction of spacetime) as the two ends of the portal through which I originally travelled forwards. If I miss by so much as a nanosecond, the children I find will only be temporal counterparts of my children. (It is not obvious that it would be reasonable for me to *care* about this mismatch, but it is nonetheless real.) For the same reasons, travel by portal cannot allow us to visit ancient Greece: the country the portal-traveller visits is not Greece, only one of its past counterparts. And portal-travel will not allow us to talk to Socrates. Socrates never lived in any country other than Greece, whereas the person named 'Socrates' with whom one will speak—a concrete past counterpart of Socrates himself—lives and always will live in a country other than Greece.

The temporal counterpart theory thus fails to classify as literally true some of the claims that people in a world of portals would naturally be tempted to make. On the other hand, there are other natural speeches that the temporal counterpart theorist can accept which are problematic from a B-theoretic point of view. For example, after the final pastwards journey of the protagonist of H.G. Wells's *The Time Machine*, the narrator remarks 'He may even now—if I may use the phrase—be wandering on some plesiosaurus-haunted Oolitic coral reef, or beside the lonely saline lakes of the Triassic Age.' Given temporal counterpart theory, 'even now' is perfectly correct. In a B-theoretic picture, by contrast, it is hard to see how the remark could be true even if one omits the 'now', given the use of the present tense.

(iii) Imagine we are at a convention of portal-travellers, who have arrived at the convention through portals whose other ends are widely scattered in spacetime. We see Zack and Maria walking along; she is dressed much more heavily than him. We wonder why are they dressed so differently. A natural response:

- (1) Just an hour ago, Maria was much colder than Zack, since she was studying mammoths in the Ice Age, whereas he was dodging dinosaurs in the Triassic.

(1) is straightforwardly true according to the temporal counterpart theory: Maria's one-hour-ago counterpart is much colder than Zack's. But claims like (1), which turn on co-ordination between disparate "personal times", cannot be accepted by B-theorists—at least, they cannot be accepted by those B-theorists who identify instants with simultaneity slices. For there may well be no simultaneity slice at which Maria is colder than Zack, on any reasonable account of what it is for one person to be colder than another at a simultaneity slice. If simultaneity slices play the 'instant' role, it follows that it is *never* the case that Maria is colder than Zack. This is manifestly inconsistent with the claim that she was colder than him an hour ago.

Could a B-theorist endorse claims like (1) by giving up the identification of instants with simultaneity slices? Perhaps. For instance, a B-theorist might identify instants with certain *functions from objects to stages*; facts about how objects are 'at' such a function, in temporally local respects such as shape and size, are determined by facts about the stages to which they are mapped by that function. In the context of an utterance that takes place in a certain simultaneity slice *s*, 'the present instant' might plausibly be taken to refer, by default, to the function that maps each object to its stage in *s*, while

'the instant one hour before the present' referred to the function that maps every object to whichever stage is one hour before its *s*-stage in *that object's personal time*.⁸ This lets us accept (1).

However, it is worth noting that B-theorists who adopt a conception of instants that works like this will find themselves having to accept many of the more surprising claims characteristic of the temporal counterpart theorist's account of portal-travel. For instance, when I travel into the future to meet my son, I might say 'An hour ago, I was walking next to you on the way to school'. But this is false if 'an hour ago' picks out a function that maps me to the stage located one hour prior to the utterance in *my* personal time and maps my son to the stage located one hour prior to the utterance in *his* personal time. (These two stages are widely separated in spacetime, and do not stand in the relation corresponding to the predicate 'next to'.) Indeed, it seems I can speak truly if I say to my son 'I have never seen or spoken to you before', since for no *n* is it the case that his *n*-seconds-before-*s* stage stands in the kind of spatiotemporal relations to mine that would be required for me to have seen or spoken to him *n* seconds ago. Similarly, if I travel through a portal that leads from modern Greece to ancient Greece, I will be able to speak truly by saying 'I have never been in this country'.

In fact, B-theorists who adopt the suggested strategy for saving (1) are worse off than temporal counterpart theorists. For while temporal counterpart theorist rejects the portal-traveller's speech 'An hour ago, I was walking next to you on the way to school', they can at least accept 'An hour ago, I was walking next to *someone* on the way to school', which is good, since this claim is obviously true in context. But on the B-theoretic strategy, the latter claim comes out false too: no person in the B-theorists' ontology has a stage one hour prior to *s* in his or her personal time which stands in the 'next to' relation to the relevant stage of the portal traveller. This is bizarre. There is no plausibility at all to the suggestion that we can accept speeches like 'One hour ago, I was feeling nervous about my impending trip by portal' but not 'One hour ago, I was walking next to my son on the way to school' as spoken by someone who has recently emerged from a portal.

B-theorists can mitigate this strangeness to some extent by appealing

⁸Probably, if we go this route, we should not require 'the present instant' *always* to refer to a function that maps each object to whichever of its stages belongs to some one simultaneity slice. For example, in the context of the the speech from *The Time Machine* quoted above, 'now' is more naturally taken to refers to a function that maps the narrator to a twentieth-century stage and the protagonist to a much earlier stage.

to context-sensitivity. They might claim that it is only in certain unusual contexts, such as that of the imagined utterance of (1), that instants of time get to track everyone's "personal time" independently. In normal contexts, such as that of the portal-traveller's conversation with his adult son, only one thing's "personal time" is relevant. In these normal contexts, 'an hour ago' might be taken to refer to the function that maps each object to whichever of its stages belongs to the same simultaneity slice as the stage of the relevant object which is one hour earlier than the utterance in the relevant object's personal time. This might help a bit, if we could tell some plausible story about why "normal" contexts should be preferred even when considerations of charity favour "unusual" contexts. But I don't think it helps that much. Claims like 'One hour ago, I was feeling nervous about my impending trip by portal but I was not walking next to anyone on the way to school' seem so crazy (in the circumstances we have been imagining) that it is hard to accept that they are true in *any* single context, even an unusual one.

(iv) Temporal counterpart theorists might also consider appealing to context-sensitivity of a similar sort, in order to avoid having to make the strange speeches discussed above under (ii). Perhaps in some contexts where a particular traveller is salient, the standards for temporal counterparthood for other objects will be set by reference to the salient traveller's personal time, rather than by their own personal times.⁹ I could then speak truly when I tell the person I meet after my futurewards trip by portal 'You are my son, and an hour ago I was walking beside you on the way to school', and similarly, before the trip I could speak truly by telling my son 'An hour from now, you will be telling me all about your life as an adult', provided that in each context I am the salient object which sets the standard for all temporal counterpart relations. My son could, in a different context where he is the salient object, say 'An hour from now, I will still be in school, and you will be nowhere at all'. We could also envisage contexts in which things work in a more complicated way, so that several different travellers each gets to play

⁹How exactly would this work? Here is one natural suggestion. Let o_1 be the salient object which sets the standard. Then for any other object o_2 , o_2 's n -seconds-hence counterpart is whichever of o_2 's temporal counterparts belongs to the same simultaneity slice as the n -seconds-hence counterpart of whichever of o_1 's temporal counterparts belongs to the same simultaneity slice as o_2 . Things get more complicated when either o_1 or o_2 has multiple temporal counterparts belonging to the same simultaneity slice. While in some cases of this sort there may be nonarbitrary grounds for choosing which to use in identifying o_2 's n -seconds hence counterpart, in others there may be no such grounds. This is another potential source of vagueness and context-sensitivity.

a role in fixing the temporal counterparts of certain other objects. We will need something like this if we want to be able to endorse generalisations like 'Since the discovery of these portals, many people have used them to visit their children as adults'.

There are two ways in which this sort of contextualist proposal could be implemented. On the one hand, we could trace the context-sensitivity to context sensitivity in the tense operators and related expressions ('temporal counterpart', '*n*-hours-earlier counterpart', 'temporal counterpairing', 'instant'). On the other hand, we could trace it to context-sensitivity in the relevant referring terms ('I', 'you', 'my son'...). On the latter proposal, the thing which the portal-traveller addresses as 'my son' is one of several coinciding objects. Among these objects, some have one-hour-earlier counterparts associated with parts of the simultaneity slice one hour earlier; others have one-hour-earlier counterparts associated with parts of the much earlier simultaneity slice that hosts the portal-traveller's one-hour-ago counterparts. In some contexts, the extension of 'person' includes objects of the former sort; in others, it includes objects of the latter sort. In each context, only one of the coinciding objects is in the extension of 'person'. The referent of terms like 'you' and 'my son' varies in such a way that 'you are a person' and 'my son is a person' is true in each context.

*** MORE TO BE FILLED IN HERE, OBVIOUSLY ***

6.4 Relativity <TK>